

Kit SDS Cover Sheet

Document ID: OSR6178-75: Version 11
Revision Date (year/month/day) 2025/06/08
Last Revision Date (year/month/day) 2024/11/02

Product information

Product name Creatinine
Part number OSR6178, OSR6678

Components

Description Creatinine R1
Creatinine R2

Transport information

Shipping information	IATA	IMDG	US DOT	European ADR	Canadian TDG
UN/ID number	1824	1824	1824	1824	1824
UN proper shipping name	Sodium hydroxide solution				
Transport hazard class(es)	8 Corrosives	8 Corrosive substances	8 Corrosive material	8 Corrosive substances	8 Corrosives
Subsidiary risk	None	None	None	None	None
Classification code	Not applicable	Not applicable	Not applicable	C5	Not applicable
Packing group	III	III	III	III	III
Special provisions	A3, A803	223	N34	None	None
Additional information					
IATA ERG code	8L	Not applicable	Not applicable	Not applicable	Not applicable
EmS	Not applicable	F-A, S-B	Not applicable	Not applicable	Not applicable
NAERG code	Not applicable	Not applicable	154	Not applicable	154

Environmental hazards

Marine pollutant	Not applicable	No	Not applicable	Not applicable	Not applicable
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Special precautions for user

Warning: Corrosive substances.



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Section 1 Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name Creatinine R1
Part number Component of P/N OSR6178, OSR6678

1.2 Relevant identified uses of the substance or mixture and uses advised against

Product use For In Vitro Diagnostic Use. See product literature for details.

1.3 Details of the supplier of the safety data sheet

Manufacturer

Beckman Coulter, Inc.
 250 S. Kraemer Blvd
 Brea, CA 92821, U.S.A.
 Tel: 800-854-3633

Supplier

CANADA
 Beckman Coulter Canada LP
 7075 Financial Drive
 Mississauga, ON L5N 6V8
 Canada
 1-800-463-7828

UNITED KINGDOM
 Beckman Coulter (UK) Ltd.
 Amersham Place
 Little Chalfont
 Buckinghamshire
 United Kingdom, HP7 9NA
 01494 441181

AUSTRALIA
 Beckman Coulter Australia Pty Ltd
 23-27 Chaplin Drive
 Lane Cove NSW 2066
 Australia
 ABN 81 002 011 672
 24 Hour emergency contact phone
 number:
 1800 060 881

SWITZERLAND
 Beckman Coulter Eurocenter SA
 22, rue Juste-Olivier, Case Postale
 1044,
 CH-1260 Nyon 1, Switzerland.
 Telephone: +41 (0)22 365 36 11
 Monday through Friday, 9:00 am to
 7:00pm)

NEW ZEALAND
 Beckman Coulter NZ
 Unit J, 33 Walmsley Road, Otahuhu,
 Auckland 1062, New Zealand
 Hours available: 08:30 - 17:00

Beckman Coulter Ireland Inc.
 Lismeehan
 O'Callaghan's Mills
 Co. Clare
 Ireland
 Tel: 353 (0)65 6831100

ICELAND / ÍSLAND
 Beckman Coulter AB
 Ekbacksvägen 28
 168 69 Bromma
 Sweden
 Phone No.: +46 80564 85 900

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Section 1 Identification of the substance/mixture and of the company/undertaking (Continued)

Hours available: 08.00-16.30

MALTA
DX Distributor:
Cherubino Ltd
DELF Building, Sliema Road, Gzira,
GZR 1637
Telephone: +356 21343270
Hours available: 08:30 – 17:00

e-mail address

SDSNT@beckman.com
Further information Contact:
Customer support Unit, Beckman Coulter Ireland Inc.
Technical Service Department Tel. +001-800-854-3633 (PST)
E-mail: Techsupportuk@beckman.com

1.4 Emergency telephone number

Telephone number (24H)

Chemtrec Emergency Tel No. U.S.A. 800-424-9300, International (001) 703-527-3887
Tel +353 (0)65 683 1170; 08:00 - 16:30 hrs Mon-Thur, 08:00 - 15:30 hrs Fri
(GMT) Tel +001-800-223-0130 (PST)

Distributor and emergency phone no.

Refer to attached list, Document ID: [472050](#), for local distributor and emergency phone numbers.

UNITED STATES - Emergency Phone (24h): Chemtrec (800) 424-9300, International (001) 703-527-3887

CANADA - Poison Centre: 1-844-764-7669; Centre antipoison du Québec: 1-800-463-5060

UNITED KINGDOM - For UK and Scotland: Emergency Call 999

IRELAND - National Poisons Information Centre Phone No.: Members of Public: +353 (01) 809 2166 (8:00 am to 10:00 pm 7 days a week); Healthcare Professionals: +353 (01) 809 2566 (24 hour service)

AUSTRALIA - 24 Hour emergency contact phone number: 1800 060 881

NEW ZEALAND - 24 Hour emergency number: 0800 446 109

Section 2 Hazards identification

2.1 Classification of the substance or mixture

Product description

In vitro diagnostic reagent.
Light yellow; Liquid; Odorless

Classification according to EC 1272/2008 (CLP/GHS)

Skin Corrosion Category 1, H314
Eye Damage Category 1, H318

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Section 2 Hazards identification (Continued)

Classification according to US-OSHA (HCS 29 CFR 1910.1200) and UN GHS

Skin Corrosion Category 1
Eye Damage Category 1

2.2 Label elements

According to EC 1272/2008 (CLP/GHS), US-OSHA and UN GHS Hazardous ingredients

Sodium Hydroxide

Pictogram



Signal word

DANGER

Hazard statements

H314 Causes severe skin burns and eye damage.

Precautionary statements

Prevention

P280 Wear protective gloves, protective clothing and eye/face protection.

Response

P301+P330+P331 IF SWALLOWED: rinse mouth. Do NOT induce vomiting.

P303+P361+P353 IF ON SKIN (or hair): Rinse skin with water.

P304+P340 IF INHALED: Remove person to fresh air and keep at rest in a position comfortable for breathing.

P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P310 Immediately call a POISON CENTER or doctor/physician.

P363 Wash contaminated clothing before reuse.

Storage

P405 Store locked up.

Disposal

P501 Dispose of contents/container in accordance with local/national regulations

Product label will display most significant precautionary statements.

2.3 Other hazards

Results of PBT and vPvB assessment

PBT: Not applicable.

vPvB: Not applicable.

See Section 11 Toxicological Information for more detailed health information.

Section 3 Composition and information on ingredients

3.2 Mixtures

Hazardous ingredients:		Hazard classification of pure ingredients		
Chemical name	% by wt.	EU 1272/2008 CLP/GHS	GHS	Note

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Section 3 Composition and information on ingredients (Continued)

Sodium Hydroxide CAS # 1310-73-2 EINECS # 215-185-5 Index # 011-002-00-6	0.5 - 1	Eye Dam. 1, H318 Skin Corr. 1A, H314 Specific Concentration Limit (SCL) Skin Irrit. 2 H315 >= 0.5% - < 2% Skin Corr. 1A H314 >= 5% Skin Corr. 1B H314 >= 2% - < 5% Eye Irrit. 2 H319 >= 0.5% - < 2%	Eye Dam. 1, H318 Skin Corr. 1A, H314	REST
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REST - Subject to Restriction as per Annex XVII of REACH Regulation (EC) No 1907/2006

See section 8 for available Occupational exposure limits

See Section 15 for additional regulatory information

See Section 16 for description of hazard class and hazard statements

Section 4 First aid measures

4.1 Description of first aid measures

Inhalation

If product is inhaled, move exposed individual to fresh air. If individual is not breathing, begin artificial respiration by trained personnel and obtain medical attention immediately.

Eye contact

If product enters eyes, rinse eyes gently with water for 15 minutes or longer, making sure that the eyelid is held open. Obtain medical advice/attention.

Skin contact

In case of skin contact, rinse with plenty of water for at least 15 minutes. Remove contaminated clothing and shoes. Obtain medical advice/attention.

Ingestion

If product is ingested, rinse mouth with water. If irritation or discomfort occurs, obtain medical attention immediately.

4.2 Most important symptoms and effects, both acute and delayed

Corrosive. May cause burns to eyes, skin and respiratory tract.

See Section 11 Toxicological Information for more detailed health information.

4.3 Indication of any immediate medical attention and special treatment needed

No further relevant information available. Refer to Section 4.1.

Section 5 Firefighting measures

5.1 Extinguishing media

In case of fire use carbon dioxide (CO₂), dry chemical, water spray or foam.
For large fires use extinguishing media suitable for surrounding fire.

5.2 Special hazards arising from the substance or mixture Special fire and explosion hazards

No special hazards determined.

Hazardous combustion products

No combustion products posing significant hazards are expected from this product (an aqueous solution).

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Section 5 Firefighting measures (Continued)

5.3 Advice for firefighters

Protective equipment Self-contained breathing apparatus is recommended for firefighters in all chemical fire situations.

Additional information No further relevant information available.

Section 6 Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions Wear protective gloves, protective clothing and tightly sealed eye/face protection. Observe general safety guidelines for protection; avoid eye and skin contact.

6.2 Environmental precautions Contain spill to prevent migration.
Do not allow the undiluted product to enter sewers/surface or ground water.

6.3 Methods and material for containment and cleaning up

Spill and leak procedures Absorb spilled material with an appropriate inert, non-flammable absorbent and dispose according to local regulations.

6.4 Reference to other sections Refer sections 8 and 13.

Section 7 Handling and storage

7.1 Precautions for safe handling Use good laboratory procedures; avoid eye and skin contact.

7.2 Conditions for safe storage, including any incompatibilities

Store at 2 to 8°C, as directed on the product label.

To maintain product quality, store according to the instructions in the product labeling.

Store away from strong acids, strong bases, strong oxidizers and incompatible materials (section 10).

7.3 Specific end uses No further relevant information available.

Section 8 Exposure controls and personal protection

8.1 Control parameters

Exposure limits

US OSHA

Sodium Hydroxide 2 mg/m3 TWA
CAS # 1310-73-2

ACGIH

Sodium Hydroxide 2 mg/m3 Ceiling
CAS # 1310-73-2

ACGIH Biological Exposure Indices (BEI)

None established

DFG MAK

None established

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Section 8 Exposure controls and personal protection (Continued)

Ireland

Sodium Hydroxide 2 mg/m3 STEL
CAS # 1310-73-2

IOELVs

None established

NIOSH

Sodium Hydroxide 10 mg/m3 IDLH
CAS # 1310-73-2

China

Sodium Hydroxide 2 mg/m3 Ceiling MAC
CAS # 1310-73-2

Croatia

Sodium Hydroxide 2 mg/m3 STEL [KGVI]
CAS # 1310-73-2

Japan

None established

Sweden (AFS 2015:7 and amendments)

Sodium Hydroxide 1 mg/m3 TLV NGV; 2 mg/m3 Binding STEL Bindande KGV
CAS # 1310-73-2

Turkey

None established

8.2 Exposure controls

Engineering controls

No special engineering controls are required. Use with good general ventilation.

Eye protection

Safety glasses or chemical goggles should be worn to prevent eye contact.

Refer U.S. OSHA 29 CFR 1910.133, European Standard EN166 or appropriate government standards.

Skin protection

Wear impervious gloves such as Nitrile or equivalent and protective clothing. Refer to U.S. OSHA 29 CFR 1910.138, European Standard EN 374, EN 14605:2005+A1:2009 or appropriate government standards.

Respiratory protection

Under normal conditions, the use of this product should not require respiratory protection. If overexposure should occur and ventilation is not adequate to maintain airborne concentrations at acceptable levels, the use of respiratory protection should be evaluated by a qualified professional.

Section 9 Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state	Liquid	Density and/or relative density	1.02
Color	Light yellow	Solubility	
Odor	Odorless	Water	Fully miscible
pH	> 13	Organic	Not determined

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Section 9 Physical and chemical properties (Continued)

Freezing point	Similar to water, approximately 0°C	Partition coefficient n-octanol/water (log value)	Not determined
Boiling point or initial boiling point and boiling range	Similar to water, approximately 100°C	Auto-ignition temp.	Product is not selfigniting
Flash point	Not applicable	Decomposition temperature	Not determined
Flammability	Not applicable	Vapor pressure	Similar to water, approximately 23 hPa
		Kinematic viscosity	Not determined
Lower and upper explosion limit	Not applicable		
Relative vapor density	Not determined		
Particle characteristics	Not applicable		
9.2 Other information			
Information with regard to physical hazard classes	Not applicable No further relevant information available.		
Other safety characteristics	No further relevant information available.		

Section 10 Stability and reactivity

10.1 Reactivity	No further relevant information available.
10.2 Chemical stability	The product is stable in accordance with recommended storage conditions.
10.3 Possibility of hazardous reactions	No further relevant information available.
10.4 Conditions to avoid	To maintain product performance keep away from strong acids, strong bases, strong oxidizers. Avoid exposure to heat and direct sunlight.
10.5 Incompatible materials	No further relevant information available.
10.6 Hazardous decomposition products	No decomposition products posing significant hazards would be expected from this product (an aqueous solution).

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Section 11 Toxicological information

11.1 Information on hazard classes

Toxicity data for hazardous ingredients

Sodium Hydroxide
CAS # 1310-73-2

Dermal LD50 Rabbit 1350 mg/kg (NLM_HSDDB); Oral LD50 Rat 325 mg/kg (OECD_SIDS)

Primary routes of exposure

Eye contact, ingestion, inhalation, and skin contact.

Acute toxicity

Not classified based on available data.

Skin corrosion/irritation

Contact may cause severe skin burns.

Serious eye damage/irritation

Contact may cause serious eye damage.

Respiratory or skin sensitisation

Not classified based on available data.

Germ cell mutagenicity

Not classified based on available data.

Carcinogenicity

No ingredients in this product are listed as carcinogens by ACGIH, IARC, NTP, OSHA or 1272/2008 EC regulation.

Reproductive toxicity

Not classified based on available data.

Specific target organ toxicity (STOT) – single exposure

Not classified based on available data.

Specific target organ toxicity (STOT) – repeated exposure

Not classified based on available data.

Aspiration hazard

Not classified based on available data.

11.2 Information on other hazards

Endocrine disrupting properties

This product does not have substance(s) of endocrine disrupting properties for health according to REACH Article 57(f).

Other information

No further relevant information available.

Section 12 Ecological information

12.1 Toxicity

Fresh water species

Sodium Hydroxide
CAS # 1310-73-2

LC50 96 h Oncorhynchus mykiss: 45.4 mg/L [static] (IUCLID)

Microtox/organisms

No information available.

Water flea

No information available.

Fresh water algae

No information available.

12.2 Persistence and degradability

Not determined for the product.

12.3 Bioaccumulative potential

Not determined for the product.

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Section 12 Ecological information (Continued)

- 12.4 Mobility in soil** Not determined for the product.
- 12.5 Results of PBT and vPvB assessment**
Not determined for the product. PBT: Not applicable, vPvB: Not applicable.
- 12.6 Endocrine disrupting properties**
This product does not have substance(s) of endocrine disrupting properties for environment according to REACH Article 57(f).
- 12.7 Other adverse effects** No further relevant information available.

Section 13 Disposal considerations

- 13.1 Waste treatment methods**
- Product waste disposal** Chemical residues and remains should be routinely handled as special waste. This must be disposed of in compliance with anti-pollution and other laws of the country concerned. To ensure compliance we recommend that you contact the relevant (local) authorities and/or an approved waste-disposal company for information.
- Package disposal** Dispose of waste product, unused product and contaminated packaging in compliance with federal, state and local regulations. If unsure of the applicable requirements, contact the authorities for information.
- Additional information** Suggested European waste catalogue 18 01 06* - chemicals consisting of or containing dangerous substances. Dispose in accordance with national, state and local waste regulations.

Section 14 Transport information

Shipping information	IATA	IMDG	US DOT	European ADR	Canadian TDG
14.1 UN/ID number	1824	1824	1824	1824	PIN - 1824
14.2 UN proper shipping name	Sodium hydroxide solution				
14.3 Transport hazard class(es)	8 Corrosives	8 Corrosive substances	8 Corrosive material	8 Corrosive substances	8 Corrosives
Subsidiary risk	None	None	None	None	None
Classification code	Not applicable	Not applicable	Not applicable	C5	Not applicable
14.4 Packing group	III	III	III	III	III
Special provisions	A3, A803	223	N34	None	None
Additional information					
IATA ERG code	8L	Not applicable	Not applicable	Not applicable	Not applicable
EmS	Not applicable	F-A, S-B	Not applicable	Not applicable	Not applicable
NAERG code	Not applicable	Not applicable	154	Not applicable	154

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Section 14 Transport information (Continued)

14.5 Environmental hazards

Marine pollutant	Not applicable	No	Not applicable	Not applicable	Not applicable
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14.6 Special precautions for user

Warning: Corrosive substances.

14.7 Maritime transport in bulk according to IMO instruments

Not applicable

Section 15 Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

US Federal and State Regulations

SARA 313 (Section 313, Title III reporting requirements)

No ingredients listed.

CERCLA (The Comprehensive Environmental Response, Compensation, and Liability Act) 40 CFR 302.4

CAS # 1310-73-2	Sodium Hydroxide
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California Proposition 65

Chemical which is known to the State of California to cause cancer

No ingredients listed.

Chemical which is known to the State of California to cause development toxicity

No ingredients listed.

Chemical which is known to the State of California to cause male reproductive toxicity

No ingredients listed.

Chemical which is known to the State of California to cause female reproductive toxicity

No ingredients listed.

Massachusetts Right To Know (RTK) List

CAS # 1310-73-2	Sodium Hydroxide
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New Jersey Dept. of Health Right To Know (RTK) List

CAS # 1310-73-2	Sodium Hydroxide
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Pennsylvania Right To Know (RTK) List

CAS # 1310-73-2	Sodium Hydroxide
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EU Regulations

This SDS complies with EC Regulations 1907/2006 (REACH) and amendments.

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Section 15 Regulatory information (Continued)

Regulation (EU) 2019/1148 on the marketing and use of explosives precursors - Substances Subject to Suspicious Transactions Reporting

No ingredients listed.

Regulation (EU) 2019/1148 on the marketing and use of explosives precursors - Restricted Explosives Precursors

No ingredients listed.

REACH 1907/2006 EC - Candidate List of Substances of Very High Concern (SVHC)

No ingredients listed.

REACH 1907/2006 EC - Annex XVII – Restrictions on Certain Dangerous Substances

CAS # 1310-73-2 Sodium Hydroxide Entry No.: 75

REACH 1907/2006 EC - Annex XIV - list of substances subject to authorisation

No ingredients listed.

Refer to Section 3

UK Regulations

UK REACH Regulation (as Amended) - List of substances subject to authorisation

Refer to Section 3

Canada

This product is exempt from WHMIS label and SDS requirements.

China

Catalog of Hazardous Chemicals - Hazardous Chemicals

CAS # 1310-73-2 Sodium Hydroxide

Inventory - China - Inventory of Existing Chemical Substances (IECSC)

All ingredients are listed or exempted.

Turkey

Türkiye-REACH - KKDIK Regulation - Annex 17 – Restrictions

No ingredients listed.

International

UN/FAO/Rotterdam Convention - Chemicals Subject to Prior Informed Consent (PIC)

No ingredients listed.

15.2 Chemical safety assessment

A Chemical Safety Assessment has not been carried out.

Some hazardous ingredients listed in Section 15 are below the cutoff limits of 0.1% for carcinogen, mutagen and reproductive toxin and 1% for other health hazards required for reporting in Section 3.

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Section 16 Other information

Beckman Coulter safety rating	Flammability: 0 Health: 3 Reactivity with water: 0 Physical contact: 3	Code 0=None 1=Slight 2=Caution 3=Severe
Revision changes	Update supplier addresses in Section 1.3	
Document version and issue/revision date	Revision Date (year/month/day) 2025/06/08 Last Revision Date (year/month/day) 2024/11/02 Document ID: OSR6178-75 Version: 11	
Hazard Classification Procedure	This mixture was classified using the calculation method for human health and environmental hazards. Physical hazards were determined based on the specification.	
Description of hazard class and hazard statements from Section 3	Eye Dam. 1 - Eye Damage Category 1 Skin Corr. 1A - Skin Corrosion Category 1A H314 - Causes severe skin burns and eye damage. H318 - Causes serious eye damage.	
Abbreviations and acronyms	ACGIH - American Conference of Governmental Industrial Hygienists (ACGIH) ADR and RID - European Agreement Concerning the International Carriage of Dangerous Goods by Road and Rail CLP - Classification, Labeling and Packaging DFGMAK - Republic Germany's maximum exposure limit EC50 - Concentration of a substance in an environmental medium expected to produce a certain effect in 50% of test organisms GHS - Globally Harmonized System of Classification and Labelling of Chemicals (GHS) HCS - Hazard Communication Standard IARC - International Agency for Research on Cancer IATA DGR - International Air Transport Association Dangerous Goods Regulation ICAO - International Civil Aviation Organization IDLH - Immediately Dangerous to Life or Health IMDG - International Maritime Dangerous Goods IMO - International Maritime Organization IOELVs - European Unions' Indicative Occupational Exposure Limit Values LC50 - Concentration of a substance in water causing death (50% of the tested population) to aquatic life LD50 - Lethal Dose 50% NIOSH - National Institute for Occupational Safety and Health NTP - National Toxicology Program	

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Section 16 Other information (Continued)

OSHA - Occupational Safety and Health Administration
PBT - Persistent Bioaccumulative and Toxic substances
PEL - Permissible Exposure Limit
SARA - Superfund Amendments and Reauthorization Act
STEL – Short Term Exposure Limit
STLV - Short Term Limit Value
STV - Short Term Value
TDG - Canadian Transportation of Dangerous Goods Regulations
TLV - Threshold Limit Value
TWA – Time Weighted Average
UN GHS - United Nations Globally Harmonized System
US DOT - United States Department of Transportation
US OSHA - United States Occupational Safety and Health Administration
vPvB - very Persistent and very Bioaccumulative substances
WHMIS - Workplace Hazardous Material Information System

For further information, please contact your local Beckman Coulter, Inc. representative.

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Section 1 Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name Creatinine R2
Part number Component of P/N OSR6178, OSR6678

1.2 Relevant identified uses of the substance or mixture and uses advised against

Product use For In Vitro Diagnostic Use. See product literature for details.

1.3 Details of the supplier of the safety data sheet

Manufacturer

Beckman Coulter, Inc.
 250 S. Kraemer Blvd
 Brea, CA 92821, U.S.A.
 Tel: 800-854-3633

Supplier

CANADA
 Beckman Coulter Canada LP
 7075 Financial Drive
 Mississauga, ON L5N 6V8
 Canada
 1-800-463-7828

UNITED KINGDOM
 Beckman Coulter (UK) Ltd.
 Amersham Place
 Little Chalfont
 Buckinghamshire
 United Kingdom, HP7 9NA
 01494 441181

AUSTRALIA
 Beckman Coulter Australia Pty Ltd
 23-27 Chaplin Drive
 Lane Cove NSW 2066
 Australia
 ABN 81 002 011 672
 24 Hour emergency contact phone
 number:
 1800 060 881

SWITZERLAND
 Beckman Coulter Eurocenter SA
 22, rue Juste-Olivier, Case Postale
 1044,
 CH-1260 Nyon 1, Switzerland.
 Telephone: +41 (0)22 365 36 11
 Monday through Friday, 9:00 am to
 7:00pm)

NEW ZEALAND
 Beckman Coulter NZ
 Unit J, 33 Walmsley Road, Otahuhu,
 Auckland 1062, New Zealand
 Hours available: 08:30 - 17:00

Beckman Coulter Ireland Inc.
 Lismeehan
 O'Callaghan's Mills
 Co. Clare
 Ireland
 Tel: 353 (0)65 6831100

ICELAND / ÍSLAND
 Beckman Coulter AB
 Ekbacksvägen 28
 168 69 Bromma
 Sweden
 Phone No.: +46 80564 85 900

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Section 1 Identification of the substance/mixture and of the company/undertaking (Continued)

Hours available: 08.00-16.30

MALTA
DX Distributor:
Cherubino Ltd
DELF Building, Sliema Road, Gzira,
GZR 1637
Telephone: +356 21343270
Hours available: 08:30 – 17:00

e-mail address

SDSNT@beckman.com
Further information Contact:
Customer support Unit, Beckman Coulter Ireland Inc.
Technical Service Department Tel. +001-800-854-3633 (PST)
E-mail: Techsupportuk@beckman.com

1.4 Emergency telephone number

Telephone number (24H)

Chemtrec Emergency Tel No. U.S.A. 800-424-9300, International (001) 703-527-3887
Tel +353 (0)65 683 1170; 08:00 - 16:30 hrs Mon-Thur, 08:00 - 15:30 hrs Fri
(GMT) Tel +001-800-223-0130 (PST)

Distributor and emergency phone no.

Refer to attached list, Document ID: [472050](#), for local distributor and emergency phone numbers.

UNITED STATES - Emergency Phone (24h): Chemtrec (800) 424-9300, International (001) 703-527-3887

CANADA - Poison Centre: 1-844-764-7669; Centre antipoison du Québec: 1-800-463-5060

UNITED KINGDOM - For UK and Scotland: Emergency Call 999

IRELAND - National Poisons Information Centre Phone No.: Members of Public: +353 (01) 809 2166 (8:00 am to 10:00 pm 7 days a week); Healthcare Professionals: +353 (01) 809 2566 (24 hour service)

AUSTRALIA - 24 Hour emergency contact phone number: 1800 060 881

NEW ZEALAND - 24 Hour emergency number: 0800 446 109

Section 2 Hazards identification

2.1 Classification of the substance or mixture

Product description

In vitro diagnostic reagent.
Yellow; Liquid; Odorless

Classification according to EC 1272/2008 (CLP/GHS)

Skin Corrosion Category 1, H314
Eye Damage Category 1, H318

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Section 2 Hazards identification (Continued)

Classification according to US-OSHA (HCS 29 CFR 1910.1200) and UN GHS

Skin Corrosion Category 1
Eye Damage Category 1

2.2 Label elements

According to EC 1272/2008 (CLP/GHS), US-OSHA and UN GHS Hazardous ingredients

Picric Acid

Pictogram



Signal word

DANGER

Hazard statements

H314 Causes severe skin burns and eye damage.

Precautionary statements

Prevention

P280 Wear protective gloves, protective clothing and eye/face protection.

Response

P301+P330+P331 IF SWALLOWED: rinse mouth. Do NOT induce vomiting.

P303+P361+P353 IF ON SKIN (or hair): Rinse skin with water.

P304+P340 IF INHALED: Remove person to fresh air and keep at rest in a position comfortable for breathing.

P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P310 Immediately call a POISON CENTER or doctor/physician.

P363 Wash contaminated clothing before reuse.

Storage

P405 Store locked up.

Disposal

P501 Dispose of contents/container in accordance with local/national regulations

Product label will display most significant precautionary statements.

2.3 Other hazards

Results of PBT and vPvB assessment

PBT: Not applicable.

vPvB: Not applicable.

See Section 11 Toxicological Information for more detailed health information.

Section 3 Composition and information on ingredients

3.2 Mixtures

Hazardous ingredients:		Hazard classification of pure ingredients		
Chemical name	% by wt.	EU 1272/2008 CLP/GHS	GHS	Note

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Section 3 Composition and information on ingredients (Continued)

Picric Acid CAS # 88-89-1 EINECS # 201-865-9 Index # 609-009-00-X	0.1 - 0.5	Acute Tox. Dermal 3, H311 Acute Tox. Inhal. 3, H331 Acute Tox. Oral 3, H301 Expl. 1.1, H201 Acute Toxicity Estimates (ATE) ATE Dermal = 300 mg/kg ATE Inhalation - Vapors = 3 mg/L ATE Oral = 200 mg/kg	Acute Tox. Dermal 3, H311 Acute Tox. Inhal. 3, H331 Acute Tox. Oral 3, H301 Expl. 1.1, H201	REST
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REST - Subject to Restriction as per Annex XVII of REACH Regulation (EC) No 1907/2006

See section 8 for available Occupational exposure limits

See Section 15 for additional regulatory information

See Section 16 for description of hazard class and hazard statements

Section 4 First aid measures

4.1 Description of first aid measures

Inhalation

If product is inhaled, move exposed individual to fresh air. If individual is not breathing, begin artificial respiration by trained personnel and obtain medical attention immediately.

Eye contact

If product enters eyes, rinse eyes gently with water for 15 minutes or longer, making sure that the eyelid is held open. Obtain medical advice/attention.

Skin contact

In case of skin contact, rinse with plenty of water for at least 15 minutes. Remove contaminated clothing and shoes. Obtain medical advice/attention.

Ingestion

If product is ingested, rinse mouth with water. If irritation or discomfort occurs, obtain medical attention immediately.

4.2 Most important symptoms and effects, both acute and delayed

Corrosive. May cause burns to eyes, skin and respiratory tract.

See Section 11 Toxicological Information for more detailed health information.

4.3 Indication of any immediate medical attention and special treatment needed

No further relevant information available. Refer to Section 4.1.

Section 5 Firefighting measures

5.1 Extinguishing media

In case of fire use carbon dioxide (CO₂), dry chemical, water spray or foam.
For large fires use extinguishing media suitable for surrounding fire.

5.2 Special hazards arising from the substance or mixture

Special fire and explosion hazards

No special hazards determined.

Hazardous combustion products

No combustion products posing significant hazards are expected from this product (an aqueous solution).

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Section 5 Firefighting measures (Continued)

5.3 Advice for firefighters

Protective equipment Self-contained breathing apparatus is recommended for firefighters in all chemical fire situations.

Additional information No further relevant information available.

Section 6 Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions Wear protective gloves, protective clothing and tightly sealed eye/face protection. Observe general safety guidelines for protection; avoid eye and skin contact.

6.2 Environmental precautions Contain spill to prevent migration.
Do not allow the undiluted product to enter sewers/surface or ground water.

6.3 Methods and material for containment and cleaning up

Spill and leak procedures Absorb spilled material with an appropriate inert, non-flammable absorbent and dispose according to local regulations.

6.4 Reference to other sections Refer sections 8 and 13.

Section 7 Handling and storage

7.1 Precautions for safe handling Use good laboratory procedures; avoid eye and skin contact.

7.2 Conditions for safe storage, including any incompatibilities

Store at 2 to 8°C, as directed on the product label.
To maintain product quality, store according to the instructions in the product labeling.
Store away from strong acids, strong bases, strong oxidizers and incompatible materials (section 10).

7.3 Specific end uses No further relevant information available.

Section 8 Exposure controls and personal protection

8.1 Control parameters

Exposure limits

US OSHA

Picric Acid
CAS # 88-89-1 0.1 mg/m³ TWA; prevent or reduce skin absorption

ACGIH

Picric Acid
CAS # 88-89-1 0.1 mg/m³ TWA

ACGIH Biological Exposure Indices (BEI)

None established

DFG MAK

Picric Acid
CAS # 88-89-1 skin notation

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Section 8 Exposure controls and personal protection (Continued)

Ireland

Picric Acid
CAS # 88-89-1 0.1 mg/m3 TWA; 0.3 mg/m3 STEL; Potential for cutaneous absorption

IOELVs

None established

NIOSH

Picric Acid
CAS # 88-89-1 75 mg/m3 IDLH; 0.3 mg/m3 STEL; 0.1 mg/m3 TWA

China

Picric Acid
CAS # 88-89-1 0.1 mg/m3 TWA

Croatia

Picric Acid
CAS # 88-89-1 0.1 mg/m3 TWA [GVI]

Japan

None established

Sweden (AFS 2015:7 and amendments)

Picric Acid
CAS # 88-89-1 0.1 mg/m3 TLV NGV

Turkey

Picric Acid
CAS # 88-89-1 0.1 mg/m3 TWA

8.2 Exposure controls

Engineering controls

No special engineering controls are required. Use with good general ventilation.

Eye protection

Safety glasses or chemical goggles should be worn to prevent eye contact.

Refer U.S. OSHA 29 CFR 1910.133, European Standard EN166 or appropriate government standards.

Skin protection

Wear impervious gloves such as Nitrile or equivalent and protective clothing. Refer to U.S. OSHA 29 CFR 1910.138, European Standard EN 374, EN 14605:2005+A1:2009 or appropriate government standards.

Respiratory protection

Under normal conditions, the use of this product should not require respiratory protection. If overexposure should occur and ventilation is not adequate to maintain airborne concentrations at acceptable levels, the use of respiratory protection should be evaluated by a qualified professional.

Section 9 Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state	Liquid	Density and/or relative density	1
Color	Yellow	Solubility	
Odor	Odorless	Water	Fully miscible

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Section 9 Physical and chemical properties (Continued)

pH	< 1	Organic	Not determined
Freezing point	Similar to water, approximately 0°C	Partition coefficient n-octanol/water (log value)	Not determined
Boiling point or initial boiling point and boiling range	Similar to water, approximately 100°C	Auto-ignition temp.	Product is not selfigniting
Flash point	Not applicable	Decomposition temperature	Not determined
Flammability	Not applicable	Vapor pressure	Similar to water, approximately 23 hPa
		Kinematic viscosity	Not determined
Lower and upper explosion limit	Not applicable		
Relative vapor density	Not determined		
Particle characteristics	Not applicable		
9.2 Other information			
Information with regard to physical hazard classes			
	Not applicable		
	No further relevant information available.		
Other safety characteristics			
	No further relevant information available.		

Section 10 Stability and reactivity

10.1 Reactivity	No further relevant information available.
10.2 Chemical stability	The product is stable in accordance with recommended storage conditions.
10.3 Possibility of hazardous reactions	No further relevant information available.
10.4 Conditions to avoid	To maintain product performance keep away from strong acids, strong bases, strong oxidizers. Avoid exposure to heat and direct sunlight.
10.5 Incompatible materials	No further relevant information available.
10.6 Hazardous decomposition products	No decomposition products posing significant hazards would be expected from this product (an aqueous solution).

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Section 11 Toxicological information

11.1 Information on hazard classes

Toxicity data for hazardous ingredients

Picric Acid
CAS # 88-89-1

Oral LD50 Rat 200 mg/kg (NZ_CCID)

Primary routes of exposure Eye contact, ingestion, inhalation, and skin contact.

Acute toxicity Not classified based on available data.

Skin corrosion/irritation Contact may cause severe skin burns.

Serious eye damage/irritation Contact may cause serious eye damage.

Respiratory or skin sensitisation Not classified based on available data.

Germ cell mutagenicity Not classified based on available data.

Carcinogenicity No ingredients in this product are listed as carcinogens by ACGIH, IARC, NTP, OSHA or 1272/2008 EC regulation.

Reproductive toxicity Not classified based on available data.

Specific target organ toxicity (STOT) – single exposure

Not classified based on available data.

Specific target organ toxicity (STOT) – repeated exposure

Not classified based on available data.

Aspiration hazard Not classified based on available data.

11.2 Information on other hazards

Endocrine disrupting properties

This product does not have substance(s) of endocrine disrupting properties for health according to REACH Article 57(f).

Other information No further relevant information available.

Section 12 Ecological information

12.1 Toxicity

Fresh water species No information available.

Microtox/organisms No information available.

Water flea No information available.

Fresh water algae No information available.

12.2 Persistence and degradability Not determined for the product.

12.3 Bioaccumulative potential Not determined for the product.

12.4 Mobility in soil Not determined for the product.

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Section 12 Ecological information (Continued)

12.5 Results of PBT and vPvB assessment

Not determined for the product. PBT: Not applicable, vPvB: Not applicable.

12.6 Endocrine disrupting properties

This product does not have substance(s) of endocrine disrupting properties for environment according to REACH Article 57(f).

12.7 Other adverse effects

No further relevant information available.

Section 13 Disposal considerations

13.1 Waste treatment methods

Product waste disposal

Chemical residues and remains should be routinely handled as special waste. This must be disposed of in compliance with anti-pollution and other laws of the country concerned. To ensure compliance we recommend that you contact the relevant (local) authorities and/or an approved waste-disposal company for information.

Package disposal

Dispose of waste product, unused product and contaminated packaging in compliance with federal, state and local regulations. If unsure of the applicable requirements, contact the authorities for information.

Additional information

Suggested European waste catalogue 18 01 06* - chemicals consisting of or containing dangerous substances. Dispose in accordance with national, state and local waste regulations.

Section 14 Transport information

Transportation of this product is not regulated under ICAO, IATA DGR, IMDG, US DOT, European ADR and RID or Canadian TDG.

14.1 UN/ID number: Not regulated for transportation

14.2 UN proper shipping name: Not regulated for transportation

14.3 Transport hazard class(es): Not regulated for transportation

14.4 Packing group: Not regulated for transportation

14.5 Environmental hazards: Not regulated for transportation

14.6 Special precautions for user: None

14.7 Maritime transport in bulk according to IMO instruments: Not applicable

Section 15 Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

US Federal and State Regulations

SARA 313 (Section 313, Title III reporting requirements)

CAS # 88-89-1

Picric Acid

1.0% de minimis concentration

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Section 15 Regulatory information (Continued)

CERCLA (The Comprehensive Environmental Response, Compensation, and Liability Act) 40 CFR 302.4

No ingredients listed.

California Proposition 65

Chemical which is known to the State of California to cause cancer

No ingredients listed.

Chemical which is known to the State of California to cause development toxicity

No ingredients listed.

Chemical which is known to the State of California to cause male reproductive toxicity

No ingredients listed.

Chemical which is known to the State of California to cause female reproductive toxicity

No ingredients listed.

Massachusetts Right To Know (RTK) List

CAS # 88-89-1 Picric Acid

New Jersey Dept. of Health Right To Know (RTK) List

CAS # 88-89-1 Picric Acid

Pennsylvania Right To Know (RTK) List

CAS # 88-89-1 Picric Acid

EU Regulations

This SDS complies with EC Regulations 1907/2006 (REACH) and amendments.

Water Hazard Class (Germany)

WGK 1, low water endangering

Regulation (EU) 2019/1148 on the marketing and use of explosives precursors - Substances Subject to Suspicious Transactions Reporting

No ingredients listed.

Regulation (EU) 2019/1148 on the marketing and use of explosives precursors - Restricted Explosives Precursors

No ingredients listed.

REACH 1907/2006 EC - Candidate List of Substances of Very High Concern (SVHC)

Not applicable.

REACH 1907/2006 EC - Annex XVII – Restrictions on Certain Dangerous Substances

CAS # 88-89-1 Picric Acid Entry No.: 75

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Section 15 Regulatory information (Continued)

REACH 1907/2006 EC - Annex XIV - list of substances subject to authorisation

Not applicable.

Refer to Section 3

UK Regulations

UK REACH Regulation (as Amended) - List of substances subject to authorisation

Refer to Section 3

Canada

This product is exempt from WHMIS label and SDS requirements.

China

Catalog of Hazardous Chemicals - Hazardous Chemicals

CAS # 88-89-1 Picric Acid

Inventory - China - Inventory of Existing Chemical Substances (IECSC)

All ingredients are listed or exempted.

Turkey

Türkiye-REACH - KKDIK Regulation - Annex 17 – Restrictions

No ingredients listed.

International

UN/FAO/Rotterdam Convention - Chemicals Subject to Prior Informed Consent (PIC)

No ingredients listed.

15.2 Chemical safety assessment

A Chemical Safety Assessment has not been carried out.

Some hazardous ingredients listed in Section 15 are below the cutoff limits of 0.1% for carcinogen, mutagen and reproductive toxin and 1% for other health hazards required for reporting in Section 3.

Section 16 Other information

Beckman Coulter safety rating	Flammability: 0 Health: 3 Reactivity with water: 0 Physical contact: 3	Code 0=None 1=Slight 2=Caution 3=Severe
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Revision changes

Update supplier addresses in Section 1.3

Document version and issue/revision date

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Section 16 Other information (Continued)

Hazard Classification Procedure This mixture was classified using the calculation method for human health and environmental hazards. Physical hazards were determined based on the specification.

Description of hazard class and hazard statements from Section 3

Acute Tox. Dermal 3 - Acute Toxicity Dermal, Category 3
Acute Tox. Inhal. 3 - Acute Toxicity Inhalation, Category 3
Acute Tox. Oral 3 - Acute Toxicity Oral, Category 3
Expl. 1.1 - Explosives, Division 1.1
H201 - Explosive; mass explosion hazard.
H301 - Toxic if swallowed.
H311 - Toxic in contact with skin.
H331 - Toxic if inhaled.

Abbreviations and acronyms

ACGIH - American Conference of Governmental Industrial Hygienists (ACGIH)
ADR and RID - European Agreement Concerning the International Carriage of Dangerous Goods by Road and Rail
CLP - Classification, Labeling and Packaging
DFGMAK - Republic Germany's maximum exposure limit
EC50 - Concentration of a substance in an environmental medium expected to produce a certain effect in 50% of test organisms
GHS - Globally Harmonized System of Classification and Labelling of Chemicals (GHS)
HCS - Hazard Communication Standard
IARC - International Agency for Research on Cancer
IATA DGR - International Air Transport Association Dangerous Goods Regulation
ICAO - International Civil Aviation Organization
IDLH - Immediately Dangerous to Life or Health
IMDG - International Maritime Dangerous Goods
IMO - International Maritime Organization
IOELVs - European Unions' Indicative Occupational Exposure Limit Values
LC50 - Concentration of a substance in water causing death (50% of the tested population) to aquatic life
LD50 - Lethal Dose 50%
NIOSH - National Institute for Occupational Safety and Health
NTP - National Toxicology Program
OSHA - Occupational Safety and Health Administration
PBT - Persistent Bioaccumulative and Toxic substances
PEL - Permissible Exposure Limit
SARA - Superfund Amendments and Reauthorization Act
STEL - Short Term Exposure Limit
STLV - Short Term Limit Value
STV - Short Term Value

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Section 16 Other information (Continued)

TDG - Canadian Transportation of Dangerous Goods Regulations
TLV - Threshold Limit Value
TWA – Time Weighted Average
UN GHS - United Nations Globally Harmonized System
US DOT - United States Department of Transportation
US OSHA - United States Occupational Safety and Health Administration
vPvB - very Persistent and very Bioaccumulative substances
WHMIS - Workplace Hazardous Material Information System

For further information, please contact your local Beckman Coulter, Inc. representative.

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